

DEMAND SYSTEMS & INCOME AND SUBSTITUTION EFFECTS

ECON101—INTRODUCTION TO MICROECONOMICS.

SPRING 2026

PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Problem 1

Suppose a consumer has utility function given by

$$u(x_1, x_2) = 2(x_1 + 10) \sqrt{\frac{x_2 + 12}{x_1 + 10}}.$$

The marginal rate of substitution (the rate at which the consumer is willing to trade good y for one extra unit of x) is given by

$$MRS = \frac{12 + x_2}{10 + x_1}.$$

Suppose that prices and income satisfy the condition $|10p_1 - 12p_2| \leq I$. The previous condition on prices and income is needed in order to obtain interior solutions.

- (a) Derive the Marshallian Demand system for this consumer.
- (b) Is good x_1 normal, inferior or neutral to income? What about good x_2 ?
- (c) Are goods x_1 and x_2 complements, substitutes or neutral to each other?
- (d) Compute the optimal consumption bundle when income is 100, $p_1 = 3$ and $p_2 = 1$.
- (e) Compute the optimal consumption bundle after the price of good 1 increases to 7, while everything else remains the same.
- (f) Suppose that the income compensation required to leave the consumer at the original utility level is 34.9086. Compute the income and substitution effects.

Problem 2

There are two goods, food and clothing, whose quantities are denoted by F and C and prices p_F and p_C respectively. There is a consumer whose income is denoted by I and her utility is given by

$$U(C, F) = \sqrt{C} + \sqrt{F}.$$

The marginal rate of substitution of clothing for food for this utility function is $MRS = \frac{\sqrt{F}}{\sqrt{C}}$.

For this exercise, it would be a good idea that you solve the utility maximization problem of the consumer by finding general expressions for the optimal levels of food and clothing as a function of prices and income, that is, find the system of demand equations

$$F(p_F, p_C, I), C(p_F, p_C, I).$$

- (a) Suppose that $I = 100$, $p_F = 1$, $p_C = 1$. What quantities does the consumer buy, and what is her resulting utility?
- (b) The framework we have been studying allows us to analyze more complicated tax schemes. Suppose that the government puts an ad-valorem tax on clothing. This means that if the consumer spends $p_C C$ on clothing, then an ad-valorem tax of τ raises expenditures on clothing to $(1 + \tau)p_C C$ so that the budget constraint becomes $(1 + \tau)p_C C + p_F F = I$. Suppose the government imposes an ad-valorem tax of 21% on clothing. Income and the price of food are the same as before. What quantities does the consumer buy and what is her resulting utility.
- (c) Decompose the price change into the substitution effect and the income effect.
- (d) What are government revenues from the 21% tax?
- (e) If the government wanted to compensate the consumer by giving a cash-transfer, how large does this cash-transfer need to be in order to restore the consumer to her old utility level?
- (f) Discussion: If you solved all the problem correctly, you should have concluded that the cash-transfer in the previous item is larger than the tax revenue from the government. Therefore, if the government tries to use the tax revenue to fund the cash-transfer it will not be enough. Think about the economic implications of this, in terms of the income and substitution effects, and in terms of how the tax is creating a distortion that, even if rebated, will still leave the consumer at a lower utility level. Think about the consequences of this discussion in terms of its real world policy implications. That is why you will hear economists say, in general, that taxes that distort prices are worse than income taxation.